We proved that if $f:[a,b]\to\mathbb{R}$ is integrable, then if we choose a partition P of $[a,b]$ consisting of n equally-spaced points and any tagging $\tau=\{x_i^*\}$ of P, then

$$\lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\Delta x=\int_a^b f(x)dx.$$

We will verify this for some simple functions $f(x)$. To do so, we will need to be comfortable with working with sums, which we now turn to.

1 Sum Formulas

We will make repeated use of the following theorem, which follows from the commutative, associative, and distributive properties of real numbers.

Theorem 1.1 (Sum Properties)**.** *If c is any constant (that is, that does not depend on i), then*

(a) $\sum_{i=m}^{n} ca_i = c\sum_{i=m}^{n} a_i$

(b) $\sum_{i=m}^{n}(a_i+b_i)=\sum_{i=m}^{n} a_i+\sum_{i=m}^{n} b_i$

(c) $\sum_{i=m}^{n}(a_i-b_i)=\sum_{i=m}^{n} a_i-\sum_{i=m}^{n} b_i$

Proof. (a) $\sum_{i=m}^{n} ca_i = ca_1+\cdots+ca_n = c(a_1+\cdots+a_n)=c\sum_{i=1}^{n} a_i$.

(b) $\sum_{i=m}^{n}(a_i+b_i)=(a_1+b_1)+\cdots+(a_n+b_n)=(a_1+\cdots+a_n)+(b_1+\cdots+b_n)=\sum_{i=m}^{n} a_i+\sum_{i=m}^{n} b_i$.

(c) Similar to (b).

□

1. Prove that $\sum_{i=1}^{n} 1 = n$.

2. Prove that $\sum_{i=1}^{n} i=\frac{n(n+1)}{2}$.
   Hint: Let $S=\sum_{i=1}^{n} i$. Write out the sum twice as

$$\begin{aligned} S &= 1+\quad 2\quad +\quad 3\quad +\cdots+(n-1)+n\\ S &= n+(n-1)+(n-2)+\cdots+\quad 2\quad +1 \end{aligned}$$

   and add the columns vertically.

3. Prove that $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$ by following the steps below.

   Let $S = \sum_{i=1}^{n} i^2$.

   (a) Write out $\sum_{i=1}^{n}[(1+i)^3 - i^3]$ to show that $\sum_{i=1}^{n}[(1+i)^3 - i^3] = n^3 + 3n^2 + 3n$.

   (b) Now rewrite $\sum_{i=1}^{n}[(1+i)^3 - i^3] = \sum_{i=1}^{n}[3i^2 + 3i + 1]$ and use Theorem 1.1 together with your answers to Problems 1 and 2 to show that $\sum_{i=1}^{n}[(1+i)^3 - i^3] = 3S + \frac{3}{2}n^2 + \frac{5}{2}n$.

   (c) Set your answers to (a) and (b) equal and solve for S.

4. Prove that $\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$ by following the steps below.

   Let $S = \sum_{i=1}^{n} i^3$.

   (a) Show that $S = \sum_{i=1}^{n} i^3 = \sum_{k=0}^{n-1} (n-k)^3$. (*Hint*: Write out the sum in reverse-order.)

   (b) Show that $\sum_{k=0}^{n-1} (n-k)^3 = n^3 + \sum_{k=1}^{n} (n-k)^3$.

   (c) Expand $(n-k)^3$ in $S = n^3 + \sum_{k=1}^{n} (n-k)^3$ and use Theorem 1.1 together with the results in Problems 1-3 to obtain an equation for S, which is readily solved to give the answer.

5. Evaluate $\sum_{i=1}^{n} i(4i^2-3)$.

2 Limits of Sequences of Sums

To calculate $\lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)$, we need to say precisely what it means for such a limit to exist. If $\lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*) = L$ for some real number L, then the sequence (a_n), where $a_n = \sum_{i=1}^{n} f(x_i^*)$ can be made arbitrarily close to L by taking n sufficiently large. That is,

$$\lim_{n\to\infty} a_n = L$$

means that, given $\varepsilon > 0$, there exists a natural number N such that $n \geq N$ implies $|a_n - L| < \varepsilon$.

When a_n is given by a formula $a_n = f(n)$, the following lemma allows us to 'replace n with a real variable x' to calculate the limit.

Lemma 2.1. *Suppose $f(x)$ is a continuous function such that $\lim_{x\to\infty} f(x) = L$, where L could be $\pm\infty$. Define a sequence (a_n) by evaluating f on the natural numbers: $a_n = f(n)$. Then $\lim_{n\to\infty} a_n = L$.*

6. Prove Lemma 2.1 or at least explain why it is true.

Example 2.2. Since $\lim_{x\to\infty} \frac{1}{x} = 0$, by Lemma 2.1 $\lim_{n\to\infty} \frac{1}{n} = 0$.

7. Find $\lim_{n\to\infty}\sum_{i=1}^{n}\frac{3}{n}\left[\left(\frac{i}{n}\right)^2+1\right]$.